PROGRAM

OPERATING SYSTEMS

Basic Fundamentals to CPU Core Fundamentals

[DATE] | [TIME]

[VENUE]

LEARNING PROGRAM FLOW

8:00 AM – 8:30 AM
30 MINUTES

REGISTRATION / ARRIVAL

- Participant arrival
- Registration and attendance
- Seating and materials
- Check computer / demo setup
- Prepare for the learning session

8:30 AM – 8:40 AM
10 MINUTES

FINAL TECHNICAL CHECK

- Check presentation and slides
- Test projector and audio
- Verify demo computer
- Prepare sample processes and CPU monitor

8:40 AM – 8:50 AM
10 MINUTES

OPENING / SCHOOL-REQUIRED OPENING RITES

- Formal opening
- School-required opening sequence
- Transition to the topic

8:50 AM – 9:00 AM
10 MINUTES

WELCOME / SESSION OBJECTIVES

- Welcome participants
- Explain the session objectives
- Introduce operating systems
- Preview the link between OS tasks and CPU cores

9:00 AM – 9:10 AM
10 MINUTES

INTRODUCTION OF THE RESOURCE SPEAKER

- Introduce the speaker
- Present the session topic
- Explain the learning sequence
- Transition to the lesson

9:10 AM – 9:50 AM
40 MINUTES

MAIN SESSION – PART 1 OPERATING SYSTEM FUNDAMENTALS

- What an operating system does
- Hardware and software interaction
- Kernel and system calls
- User mode and kernel mode
- Processes and threads
- Memory management basics
- Files and device management
- CPU scheduling overview

CONTINUES ON PAGE 2

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CONTINUATION / DETAILED PROGRAM FLOW

9:50 AM – 9:57 AM
7 MINUTES

PAKULO SA PROGRAM #1: OS QUICK QUIZ

CPU Core Quiz
- Quick concept questions
- Identify OS responsibilities
- Recap key terms

9:57 AM – 10:17 AM
20 MINUTES

GUIDED DISCUSSION / Q&A

- Clarify key concepts
- Discuss process and thread examples
- Ask practical CPU questions

10:17 AM – 10:27 AM
10 MINUTES

INTERMISSION NUMBER

- Introduce performance
- Short performance / intermission
- Audience energy reset
- Return to technical session

10:27 AM – 10:32 AM
5 MINUTES

TECHNICAL RESET / DEMO SETUP

- Restore presentation
- Prepare Task Manager / system monitor
- Check sample workload
- Refocus audience
- Transition to Part 2

10:32 AM – 11:12 AM
40 MINUTES

CPU CORES, SCHEDULING, AND PERFORMANCE

MAIN SESSION – PART 2

Processes

- A process is a running program
- Threads are execution paths
- The OS schedules runnable threads
- CPU usage changes with workload

THREADS

CPU Cores

- OS assigns CPU time
- Scheduling affects responsiveness
- Context switching has overhead
- Compare single-core and multi-core
- Understand parallel work
- Read CPU utilization
- Identify workload limits

CPU SCHEDULING

- Multiple cores can run tasks in parallel
- Threads may share process resources
- More cores do not guarantee every app is faster
- CPU scheduling basics
- Core utilization
- Task Manager interpretation
- Performance and thermal limits

11:12 AM – 11:19 AM
7 MINUTES

PAKULO SA PROGRAM #2

- View a process / CPU usage example
- Predict what the OS will do
- Compare single-core and multi-core
- Audience participation
- Explain the result

AUDIENCE CPU CHALLENGE

11:19 AM – 11:39 AM
20 MINUTES

OPERATING SYSTEM / CPU CORE DEMO

- Open Task Manager or system monitor
- Inspect process and CPU usage
- Compare activity across cores
- Discuss scheduling and workload
- Explain observations

11:39 AM – 11:54 AM
15 MINUTES

FINAL Q&A / KEY TAKEAWAYS

- Final questions
- Review OS responsibilities
- Recap processes, threads, and cores
- Practical performance reminders

10 MINUTES

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OPERATING SYSTEMS: CPU CORE FUNDAMENTALS

Basic Fundamentals to CPU Core Fundamentals

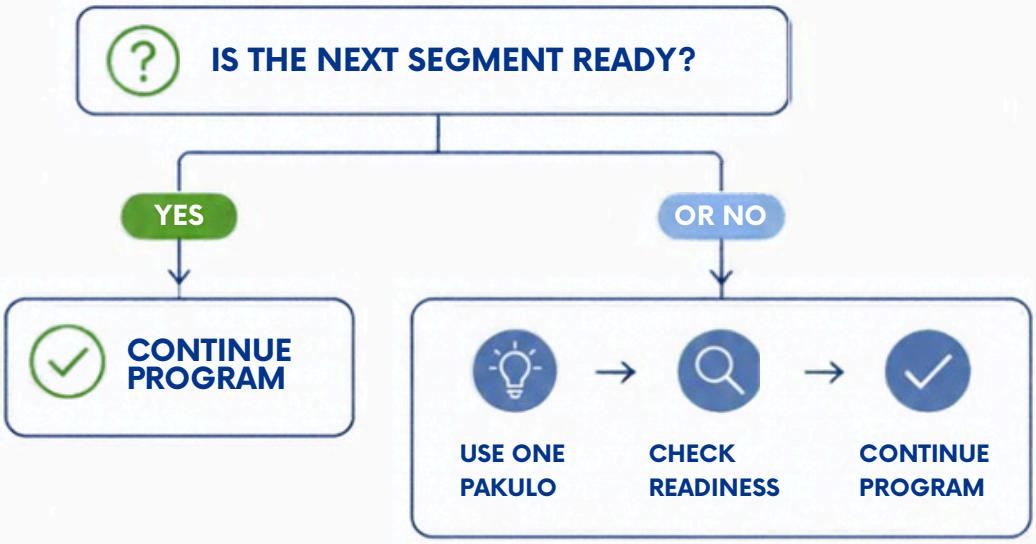
[DATE] | [TIME]

[VENUE]

PAKULO SA PROGRAM

Short activities to reinforce concepts, check understanding, and keep the technical session engaging.

- **A — OS QUICK QUIZ (3–7 MINUTES)**
- **B — CPU CORE CHALLENGE (3–7 MINUTES)**
- **C — EXPLAIN THE CONCEPT (3–5 MINUTES)**
- **D — CONTROLLED TECHNICAL RESET (UP TO 5 MINUTES)**



PAKULO is a short learning activity within the program flow, not a separate event.

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[DATE] | [TIME]
[VENUE]



10:17 AM – 10:27 AM
INTERMISSION NUMBER

10 MINUTES



DEMO / TECHNICAL TEAM



10:27 AM – 10:32 AM
TECHNICAL RESET / DEMO SETUP

5 MINUTES

- Restore presentation
- Prepare system monitor
- Reset demo workload
- Refocus audience



10:32 AM – 11:12 AM
MAIN SESSION – PART 2

40 MINUTES

CPU CORES, SCHEDULING, AND PERFORMANCE

PROCESSES

- Processes and threads
- How CPU time is assigned
- Context switching
- System responsiveness

THREADS

- Single-core vs multi-core
- Parallel execution
- CPU utilization
- Workload limits
- Performance considerations

CPU CORE FUNDAMENTALS

- Observe CPU usage
- Compare core activity
- Discuss scheduling
- Explain performance differences

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WHAT HAPPENS NEXT?

LEARNING PROGRAM FLOW

01 REGISTRATION

02 OPENING

03 WELCOME & OBJECTIVES

04 RESOURCE SPEAKER INTRODUCTION

05 OS FUNDAMENTALS

06 PAKULO

07 Q&A

08 INTERMISSION NUMBER

09 TECHNICAL RESET

10 CPU CORE FUNDAMENTALS

11 PAKULO / QUIZ

12 PRACTICAL CPU DEMO

13 FINAL Q&A

14 CLOSING REMARKS

15 GROUP PHOTO